Lösung: Symmetrie von Funktionen

$f(-x) = f(x)$	$\Leftrightarrow$	gerade Symm.
$f(-x) = -f(x)$	$\Leftrightarrow$	ungerade Symm.

a) $f(x) = \sin(x^2)$

$f(-x) = \sin((-x)^2) = \sin(x^2) = f(x)$

$\Leftrightarrow$ gerade Symmetrie

b) $f(x) = \cos^3(x) = (\cos(x))^3$

$f(-x) = (\cos(-x))^3 = (\cos(x))^3 = f(x)$

$\Leftrightarrow$ gerade Symmetrie

c) $f(x) = 5x^5 - 2x^3$

$f(-x) = 5(-x)^5 - 2(-x)^3 = -5x^5 + 2x^3$

$= -(5x^5 - 2x^3) = -f(x)$

$\Leftrightarrow$ ungerade Symmetrie

d) $f(x) = \sin(x) \cdot \cos(x)$

$f(-x) = \sin(-x) \cdot \cos(-x) = -\sin(x)\cos(x) = -f(x)$

$\Leftrightarrow$ ungerade Symmetrie

e) $f(x) = \frac{x^2 + 2}{x + \sin(x)}$

$$f(-x) = \frac{(-x)^2 + 2}{(-x) + \sin(-x)} = \frac{x^2 + 2}{-x - \sin(x)}$$

$$= \frac{x^2 + 2}{-(x + \sin(x))} = -\frac{x^2 + 2}{x + \sin(x)} = -f(x)$$

$\Leftrightarrow$ ungerade Symmetrie